「Norm and Trace.

「Def. Let K be a finite extension of F, and let $\alpha \in K$.
Let L be a Galois extension of F containing K, and let $H \leq Gal(L/F)$ corresponding to K.
Define
$$N_{K/F} : K \to F ; \alpha \mapsto \prod_{\sigma \in S} \sigma(\alpha) \quad \text{where } S = \{\tau : K \to \overline{F} \mid \tau \text{ is embedding}\}$$
and
$$Tr_{K/F} : K \to F ; \alpha \mapsto \sum_{\sigma \in S} \sigma(\alpha) \quad //$$
In particular, if K is Normal Extension over F, then $S = Gal(K/F)$.

「Prop. To simplify, we assume that K/F is Normal. Then,
1). The Norm and Trace are well-defined.
2). The Norm preserves multiplication, and
The Trace preserves addition.
3). For $\alpha \in K$, suppose that $m_\alpha(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0 \in F[x]$ is the minimal of α.
Then, d divides $[K:F] = n$,
$$N_{K/F}(\alpha) = (-1)^n \cdot a_0^{\frac{n}{d}} \quad \text{and} \quad Tr_{K/F}(\alpha) = -\frac{n}{d} \cdot a_{d-1}.$$

Proof. Given $\tau \in G(K/F)$, and $\alpha \in K$,
$$\tau(N_{K/F}(\alpha)) = \tau\Big(\prod_{\sigma \in Gal(K/F)} \sigma(\alpha)\Big) = \prod_{\sigma \in Gal(K/F)} (\tau \circ \sigma)(\alpha) \underset{\text{Since } \tau, \sigma \text{ are Automorphism}}{=} \prod_{\sigma \in Gal(K/F)} \sigma(\alpha) = N_{K/F}(\alpha),$$
So the norm is fixed by $G(K/F)$, it means the norm is contained in F, Similarly trace.
And, given $\alpha, \beta \in K$,
$$N_{K/F}(\alpha\beta) = \prod_{\sigma \in Gal(K/F)} \sigma(\alpha\beta) = \Big(\prod \sigma(\alpha)\Big) \cdot \Big(\prod \sigma(\beta)\Big) = N_{K/F}(\alpha) \cdot N_{K/F}(\beta), \quad \text{Similarly, trace.}$$
Now, let $m_\alpha(x)$ be the minimal polynomial of α. Then,
$[K:F(\alpha)] \cdot [F(\alpha):F] = [K:F] = n$, and $[F(\alpha):F] = \deg m_\alpha = d$, so $d \mid n$.
So, $|Gal(K/F)| = n$, and $\sigma \in Gal(K/F)$ conjugates α to root of m_α, therefore there are n/d repetation in the product. Consequently,
$$N_{K/F}(\alpha) = (-1)^n \cdot a_0^{n/d}.$$
And,
$$Tr_{K/F}(\alpha) = -\frac{n}{d} \cdot a_{d-1}$$
(observe $m_\alpha(x) = (x - \alpha_1) \cdots (x - \alpha_d)$).

「Example. For $K = \mathbb{Q}(\sqrt[4]{2}, i)$, find a Norm of $1 + \sqrt[4]{2} + i$.

Proof. Since $\mathbb{Q}$ is perfect, K is separable. And, a polynomial $x^4 - 2$ has four roots

$$\sqrt[4]{2},\ \sqrt[4]{2}i,\ -\sqrt[4]{2},\ -\sqrt[4]{2}\cdot i.$$

These are all contained in K, and the splitting field of x^4-2 must contain

$$\sqrt[4]{2} \text{ and } \sqrt[4]{2}i/\sqrt[4]{2} = i,$$

so K is finite Normal Extension of $\mathbb{Q}$, so $Gal(K/F)$ is the Galois group.

Since $\sigma \in Gal(K/\mathbb{Q})$ maps i to $\pm i$,

$\sqrt[4]{2}$ to $\sqrt[4]{2}\cdot i^k$ $(k = 0,1,2,3)$, so, the norm is

$$(1+i+\sqrt[4]{2})(1+i-\sqrt[4]{2})(1+i+\sqrt[4]{2}i)(1+i-\sqrt[4]{2}i)(1-i+\sqrt[4]{2})(1-i-\sqrt[4]{2})(1-i+\sqrt[4]{2}i)(1-i-\sqrt[4]{2}i)$$

$$= ((1+i)^2 - \sqrt{2})((1+i)^2 + \sqrt{2})((1-i)^2 - \sqrt{2})((1-i)^2 + \sqrt{2})$$

$$= ((1+i)^4 - 2)((1-i)^4 - 2)$$

$$= (1+i)^4 \cdot (1-i)^4 - 2((1+i)^4 + (1-i)^4) + 4$$

$$= 2^4 + 4 - 2(\underbrace{(1+i)^4}_{-4} + \underbrace{(1-i)^4}_{-4})$$

$$= 2^4 + 4 + 16$$

$$= 36.$$

$1 + 2i - 1$
$= 2i$
$4i^2 = -4$

$1 - 2i - 1$
$= -2i$
-4

Let K/F be an Extension field of degree n, and let

$$\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$$

be a basis for K over F.

Prop. K is isomorphic to a subfield of $M_{n\times n}(F)$.

Proof. Consider a map $K \to \mathcal{L}(K,K) : \alpha \mapsto T_\alpha$

where $T_\alpha(x) = \alpha \cdot x$. By simple calculation, T_α is linear, so this map is well-defined. Moreover, this map is injective linear map, because: given $\alpha, \beta \in K$ and $c \in F$,

$$T_{c\alpha+\beta}(x) = (c\alpha+\beta)\cdot x = c\alpha x + \beta x = c\cdot T_\alpha(x) + T_\beta(x).$$

And, $T_\alpha \equiv 0$ iff $\alpha = 0$, so kernel of this map is trivial, so injective. Relates $\mathcal{L}(K,K)$ and $M_{n\times n}(F)$ is omitted.

$(1-t)^3 - 2(1-t) - 2(1-t) + 2t$

Prop. The characteristic polynomial of $[T_\alpha]_\mathcal{B}$ has a root as α.

Proof. Let $\alpha \in K$ be fixed, with $\alpha \neq 0$. Put $A = [T_\alpha]_\beta$.

$6t$

By definition, $T_\alpha(\alpha_j) = \alpha\cdot\alpha_j = \sum_{i=1}^{n} A_{ij}\cdot\alpha_j$.

$(1-t)^3 = 1 - t^3 - 3t(1-t)$

$= 1 - t^3 + 3t^2 - 3t$

Let $V = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in K^n$. Then, $A^t V = \begin{pmatrix} \alpha\cdot\alpha_1 \\ \vdots \\ \alpha\cdot\alpha_n \end{pmatrix} = \alpha\cdot V$

or $(\alpha I - A^t)V = 0$. since $V \neq 0$, $\det(\alpha I - A^t) = 0$.

Thus proved. $\boxtimes$

$= (1-t)^3 - 2(1-t) - 2(1-t) + 4$

$- 2(1-t) + 6$

Example. Find a monic minimal polynomial satisfied by $\sqrt[3]{2}$ and $1 + \sqrt[3]{2} + \sqrt[3]{4}$.

Let $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$ be the basis for $\mathbb{Q}(\sqrt[3]{2})$ over $\mathbb{Q}$.

Put $\alpha = 1 + \sqrt[3]{2} + \sqrt[3]{4}$. Then,

$$T_\alpha(1) = 1 + \sqrt[3]{2} + \sqrt[3]{4},\quad T_\alpha(\sqrt[3]{2}) = 2 + \sqrt[3]{2} + \sqrt[3]{4},\quad T_\alpha(\sqrt[3]{4}) = 2 + 2\cdot\sqrt[3]{2} + \sqrt[3]{4}$$

So, $[T_\alpha]_\mathcal{B} = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$, and $\det([T_\alpha]_\beta - tI) = -t^3 + 3t^2 + 3t + 1$